

Ovarian Malignance Risk Prediction from Medical Notes: a case study in Vietnam

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Abstract—Timely and accurate assessment of ovarian tumor malignancy is critical for early intervention and effective treatment planning. This study investigates the effectiveness of artificial intelligence (AI) models in predicting malignancy risk based on Vietnamese ultrasound medical reports. A real-world dataset of 805 patients from the National Hospital of Obstetrics and Gynecology was collected. Textual data were extracted using an Optical Character Recognition (OCR) technique and processed, then regressed by various machine learning algorithms. Besides, to enhance the performance of models, we extracted key clinical findings and represented them in a tabular structure. ADASYN oversampling was applied to address class imbalance. Experimental results show that models using structured data, particularly RF, significantly outperform free text approaches, achieving an MSE of 0.032, R^2 of 0.609, and accuracy of binary classification of 93.2%. This case study demonstrates the effectiveness of combining structured AI pipelines with clinical domain knowledge for ovarian cancer risk stratification in low-resource settings.

Index Terms—Ovarian cancer, malignancy prediction, medical notes, ultrasound report, machine learning, deep learning

I. INTRODUCTION

Ovarian cancer is the 18th most common cancer worldwide and ranks as the 8th most common cancer in women, as reported by the World Cancer Research Fund. In Vietnam, the situation is equally alarming. According to the 2020 statistics from the Surveillance, Epidemiology, and End Results (SEER) Program [1], there are approximately 1,404 new ovarian cancer cases and 923 deaths annually. The disease is classified into four stages (FIGO I–IV), with corresponding 5-year survival rates of 89%, 71%, 41%, and 20%, respectively, as reported by the AJCC 6th edition [2]. These numbers underscore the urgent need for early and accurate prediction of malignancy risk to improve patient outcomes and reduce mortality.

In clinical practice, ultrasound is the most common first imaging tool used for ovarian tumor evaluation. Radiologists typically describe tumor features in semi-structured medical notes, which are then used to assess the likelihood of malignancy. However, the variability in language, terminology, and formatting can be subjective and varies across institutions and clinicians, which makes manual interpretation time-consuming and prone to subjectivity. To address this challenge, AI-based models have emerged as promising tools for malignancy risk prediction, offering the potential for consistent, scalable, and accurate assessment based on clinical data.

One of the most widely validated diagnostic frameworks for ovarian tumor assessment is the ADNEX model developed by the International Ovarian Tumor Analysis (IOTA) group. This model utilizes logistic regression applied to structured ultrasound features [3]. Alongside other protocols such as O-RADS and LR2, the ADNEX model has demonstrated high diagnostic accuracy and sensitivity across multiple countries and clinical settings [4]. However, despite their effectiveness, these models depend heavily on standardized, structured inputs, which limits their applicability in real-world environments where semi-structured or unstructured clinical text is more commonly encountered.

Building upon this foundation, our study aims to systematically evaluate whether advanced machine learning (ML) and deep learning (DL) methods can achieve comparable or improved performance relative to the traditional IOTA-based models in the classification of ovarian masses. Rather than replacing IOTA, our approach focuses on benchmarking these modern techniques against it, particularly using clinical data collected from Vietnamese patients. This evaluation provides insights into the generalizability and potential advantages of ML/DL methods when applied to localized, resource-constrained clinical settings.

In this study, we propose an end-to-end AI pipeline that begins from real-world ultrasound report images collected from 805 Vietnamese patients with ovarian or adnexal masses. Our pipeline integrates OCR, text extraction via LLM, and feature transformation into tabular formats, followed by training of multiple regression models. A particular emphasis is placed on handling imbalanced data distributions via Adaptive Synthetic Sampling - ADASYN¹, and investigating the impact of synthetic oversampling on diagnostic performance. Our contributions are three-fold. First, we introduce a structured dataset of Vietnamese medical reports, which provides a valuable resource for initiating and validating ML approaches in ovarian tumor prediction. Second, we propose a comprehensive pipeline for data preprocessing, feature extraction, and model development to facilitate systematic evaluation. Third, we conduct extensive experiments with various ML/DL models and provide a comparative analysis to identify the most effective approach, serving as a reference for future research.

¹<https://github.com/stavskal/ADASYN>

II. RELATED WORK

A. Ovarian Cancer Risk Classification Methods

Numerous studies have focused on developing predictive models to assess the malignancy risk of ovarian tumors, primarily using imaging, biomarkers, and clinical features. Traditional diagnostic scoring systems such as the Risk of Malignancy Index [5] and The International Ovarian Tumor Analysis Rules [3] combine ultrasound findings with serum biomarkers CA-125 and menopausal status to guide clinical decision making. While effective in many cases, these rule-based methods are often limited by their dependency on manually selected parameters and threshold tuning.

1) *Risk of Malignancy Index - RMI*: RMI is one of the most widely used scoring systems for assessing the likelihood of ovarian cancer. Originally proposed by Jacobs et al., the RMI combines three key factors: ultrasound findings, serum CA-125 levels, and menopausal status into a single score [5]. The simplicity and interpretability of RMI have made it a preferred tool in many clinical settings, particularly in resource-limited environments. However, studies have reported that RMI performance varies across populations.

2) *The International Ovarian Tumor Analysis - IOTA*: IOTA [3] offers a standardized method for classifying adnexal masses via ultrasound, using features like tumor size, morphology, and Doppler signals. The most commonly used IOTA-based tool is the ADNEX model (Assessment of Different NEoplasias in the adneXa), which classify ovarian tumors into benign or malignant categories. The IOTA ADNEX models, based on logistic regression of ultrasound features, have shown strong diagnostic performance across diverse settings. Validation studies include cohorts in China (n=278, 144) [6], [7], Brazil (n=876, AUC 0.880 with Modified Benign Descriptors; AUC 0.886 for O-RADS) [8], and another Brazilian study reporting up to 95% accuracy [9]. In Southeast Asia, LR2 was validated in a Singaporean cohort (n=353), supporting its generalizability [10].

B. Datasets for Ovarian Cancer Prediction

Access to ovarian cancer datasets remains limited, hindering AI development and reproducibility. Key datasets like IOTA [11] and UKCTOCS [12] require formal approval and have ethical or regional restrictions. While SEER [1] is publicly available, it lacks detailed clinical data essential for AI modeling. The scarcity of accessible, clinically rich datasets, especially from Southeast Asia, limits equitable AI research in gynecologic oncology. To address this, our study introduces an AI pipeline using 805 real-world ovarian ultrasound reports from Vietnamese patients.

III. MATERIALS AND METHODS

Our framework for predicting the benign probability or malignancy risk and classifying ovarian tumors (benign vs. malignant) is illustrated in Figure 1. In the following, we will describe the collected datasets and the methodologies.

A. OvaNOTE Dataset

In this work, we introduce **OvaNOTE**, a novel dataset of 805 de-identified ultrasound report images retrospectively collected between 2023 and 2025 from the National Hospital of Obstetrics and Gynecology in Vietnam. Due to strict data privacy policies and inaccessibility of hospital information systems, direct digital records were not available. As a result, the dataset was constructed from photographed paper-based reports under controlled conditions. These reports, routinely used by sonographers, contain textual descriptions of ovaries, uterus, and abdominal cavity, including tumor-related features such as lesion size, wall structure, multilocularity, echogenicity, papillary projections, vascularity, and ascites, written in Vietnamese by expert radiologists.

To ensure data consistency and relevance, we applied exclusion criteria: removing records without ovarian descriptions, missing IOTA-based annotations, duplicate cases, or images with poor legibility. Before any automated processing, we manually removed all personally identifiable information, including patient name, age, ID number, address, and department code, directly from the image files. All images were then jointly reviewed by clinical experts and AI engineers to ensure complete anonymization and semantic validity prior to downstream use. This approach reflects the real-world data challenges in under-resourced healthcare systems, where only paper-based records are available, and access to digital information systems is highly restricted.

To prepare the data for learning tasks, we employed two complementary feature extraction approaches:

- 1) **OvaNOTE1** (Free-text representation): The complete clinical description was preserved in Vietnamese as an unstructured natural language, enabling NLP-based modeling.
- 2) **OvaNOTE2** (Structured feature extraction): We manually parsed standardized IOTA-relevant descriptors (e.g., wall irregularity, multilocularity) into tabular form.

B. OvaNOTE1

After anonymization, we applied the Tesseract OCR¹ engine (configured for Vietnamese language support) to convert each scanned report into raw text. Given the diversity in document quality and formatting, additional post-OCR cleaning was necessary to normalize encoding, remove stray characters, separate line items logically and check medical spelling. To identify relevant segments, we built a custom ovarian-specific keyword dictionary in collaboration with clinical experts. This enabled us to extract sections describing adnexal masses and associated clinical notes. The cleaned text corpus formed the OvaNOTE1 subset, suitable for NLP-based prediction tasks. We constructed a token vocabulary of 1671 terms from the corpus and generated TF-IDF features. The Benign Probability (used as a regression/classification label) was derived directly from the IOTA model prediction provided in the medical notes.

¹<https://github.com/tesseract-ocr/tesseract>

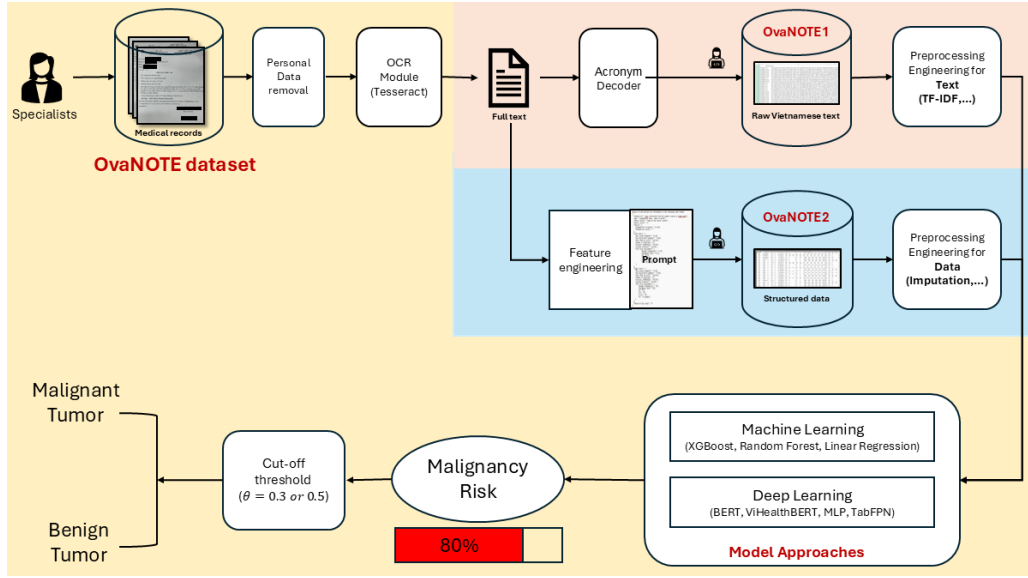


Fig. 1. Overview of the proposed framework for ovarian tumor prediction.

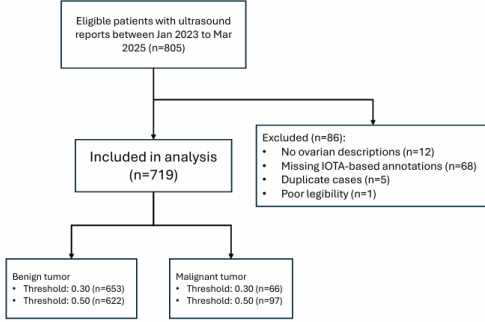


Fig. 2. Flow diagram of patient inclusion and threshold-based labeling.

Figure 2 illustrates the enrollment and data preprocessing pipeline for women diagnosed with adnexal masses between January 2023 and March 2025. From 806 records initially collected patients, data were split into two branches: one excluded cases without benign probability or missing information ($n=86$), while the other retained patients with valid benign probability scores ($n=719$).

C. OvaNOTE2

Each medical note was processed using a GPT-4.0 model, which was prompted to extract relevant clinical variables and structure them into JSON format. The structured output included variables such as the presence of solid components, multilocularity, vascularization, ascites, wall irregularity, and tumor dimensions. Each variable was converted into a binary or numerical format to enable tabular representation. Specifically, "Yes", "No", and "Unknown" were encoded as 1, 0, and NaN, respectively. Only patients exhibiting solid nodules with corresponding IOTA-derived benign probabilities were retained for analysis. To validate the accuracy of extracted

information, each JSON output was manually reviewed against the original medical note by a human annotator. This verification step ensured that each extracted value was consistent with the source report. After applying normalization rules and excluding non-informative discrepancies, the final evaluation yielded an error rate of 18.93%, corresponding to 4,935 mismatches across 26,070 compared fields. This result highlights the strong performance of the LLM-based extraction pipeline and supports its applicability for downstream machine learning tasks.

For missing data, simple imputation strategies were applied: numerical features were filled with the mean of the training set, while binary features were set to 0 where clinically justifiable as "not observed". This process produced a clean CSV file - OvaNOTE2 for each patient record, ready for ingestion by the ML/DL model. Unlike in OvaNOTE1, where data was processed per patient, OvaNOTE2 treats each ovary (left or right) as an individual sample. Thus, if a patient has descriptions for both ovaries, each side is considered separately, with its own set of features and IOTA-assigned label. For patients with both left and right ovary evaluations, we selected the lower of the two IOTA Benign Probabilities to reflect the worst-case malignancy risk. For label construction, we leveraged IOTA ADNEX risk values and mapped malignancy probabilities to classification based on various cut-off thresholds. After filtering for valid structured attributes and IOTA probabilities, the final dataset comprised 820 records, including 387 for the left side and 433 for the right.

Each record included ten key features: Age, Pelvic Fluid, Endometrial Thickness, Endometrial Cavity Status, Lesion Diameter, Solid Diameter, > 10 Cysts, Papillae, Acoustic Shadowing, and Ascites. They are collected following IOTA rules and with the suggestion from clinical experts. A statistical summary of these features is presented in Table I, grouped

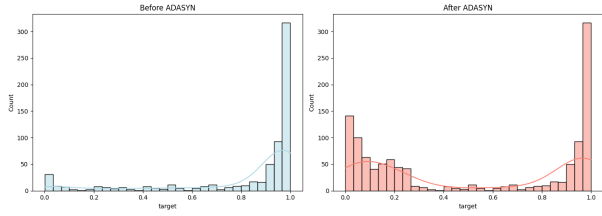


Fig. 3. Histogram of Benign Probability before/after ADASYN in one fold

TABLE I
DESCRIPTIVE STATISTICS OF STRUCTURED FEATURES IN OVA NOTE2

Feature Type	Feature Name	Mean $\pm$ deviation	Median [min; max]
Numerical	Age (years)	37.2 $\pm$ 15.2	36.5 [10.0; 88.0]
	Endometrial Thickness (mm)	7.8 $\pm$ 3.5	7.8 [0.0; 27.0]
	Lesion Diameter (mm)	94.6 $\pm$ 54.3	81.5 [0.0; 321.0]
	Solid Diameter (mm)	14.5 $\pm$ 28.0	0.0 [0.0; 210.0]
	Papillae (number)	0.4 $\pm$ 1.4	0.0 [0.0; 11.0]
	Benign Probability (%)	0.8 $\pm$ 0.3	1.0 [0.0; 1.0]
Feature Type	Feature Name	No	Yes
Categorical	Pelvic Fluid	0.779 (n=639)	0.221 (n=181)
	Abnormal endometrial cavity	0.574 (n=471)	0.426 (n=349)
	>10 Cysts	0.804 (n=658)	0.161 (n=161)
	Acoustic Shadowing	0.705 (n=578)	0.295 (n=242)
	Ascites	0.570 (n=467)	0.430 (n=353)

into numerical and categorical variables. This table highlights the underlying clinical distribution and guides subsequent model development for malignancy prediction for the entire dataset of 820 cases in OvaNOTE2.

Class distribution is summarized in Fig. 3 by blue color. Such an imbalance can bias the model toward overpredicting the majority of benign class, leading to suboptimal performance on malignant cases.

D. Prediction Approaches

In this work, we conduct two main tasks: regression and classification. The regression task aimed to predict the probability (ranging from 0 to 1) that a patient was benign (as high probability) or malignant (as low probability), while the classification task binarized predictions using varying thresholds to identify benign vs. malignant categorization. In particular, we try to answer our two research questions:

- 1) Q1 - Whether conventional/large language models classify benign/malignant tumors directly from medical text?
- 2) Q2 - What is the model that best fits the structured data?

1) *Data preprocessing*: For **OvaNOTE1**, preprocessing involved converting numerals to words, expanding medical abbreviations via a domain-specific dictionary, standardizing terminology, tokenizing, removing stopwords, and lemmatizing. Texts were then vectorized using TF-IDF. For **OvaNOTE2**, categorical characteristics were labeled, continuous characteristics were normalized to [0,1], and missing values were imputed using clinically informed defaults. Both data sets were split 80:20 and evaluated using 5-fold stratified cross-validation. To mitigate class imbalance, ADASYN was applied only in the training set, generating synthetic high-risk samples to balance the class distribution without introducing data leakage. The dataset after ADASYN is shown in Figure 3 in red color.

2) *Language models for malignancy prediction*: For free-text data in OvaNOTE1, we evaluated ML/DL-based language models to predict benign probability or malignancy risk. Initially, textual descriptions were preprocessed and transformed into TF-IDF feature vectors using a curated, domain-specific vocabulary. These vectors were then used to train classical ML regressors, including LinearR, RF, and XGBoost. We also employed pretrained transformer-based language models, namely BERT² and ViHealthBERT³, to perform direct benign probability prediction from raw clinical text. These models were fine-tuned on our dataset using regression objectives. While BERT was pretrained on general English corpora, ViHealthBERT was pretrained on Vietnamese biomedical texts, making it more suitable for our bilingual medical notes.

3) *Structured data-based prediction/classification models*: Using structured data (from OvaNOTE2), we deploy various ML models, including LinearR, RF, and XGBoost. These models are provided in the sklearn library. Besides, we also deploy DL models such as TabPFN (Tabular Prior-Data Fine-Tuned Network). TabPFN is a transformer-based model specifically pre-trained on a broad collection of synthetic and real tabular datasets⁴. TabPFN predicts outputs using Bayesian posterior inference, enabling zero-shot generalization and fast deployment in low-data regimes.

IV. EXPERIMENTS

A. Experiment Setup

All experiments were conducted in Python using scikit-learn, XGBoost, and PyTorch libraries on Google Colab with a standard GPU runtime.

The regression task aimed to predict the continuous benign probability from IOTA scores (ranging from 0 to 1), while the classification task involved threshold-based binarization. Although thresholds between 0.3 and 0.5 are commonly used in clinical practice, we adopted a 0.5 cut-off to evaluate its impact on classification accuracy (see Fig. 4). We applied 5-fold stratified cross-validation to preserve malignancy distribution across folds. For classification, we report Accuracy, Precision, Recall, and AUC; for regression, we use R^2 , MSE, and MAE. All reported results are averaged across the five folds.

B. Classification evaluation on OvaNOTE1 dataset

In this subsection, we conduct experimentation to evaluate the performance of the classification models using the free text dataset OvaNOTE1. We utilized with 0.5 cut-off threshold to decide whether a benign or malignant patients. It answers the first question: **Q1 - Whether conventional/deep models classify benign/malignant tumors directly from medical text?**

Table II presents the prediction performance of various models on the OvaNOTE1 dataset with 0.5-threshold using 5-fold cross-validation. Results on the OvaNOTE1 dataset show that both traditional ML models and large language models

²<https://huggingface.co/distilbert/distilbert-base-uncased>

³<https://huggingface.co/demdecuong/vihealthbert-base-word>

⁴<https://github.com/PriorLabs/TabPFN>

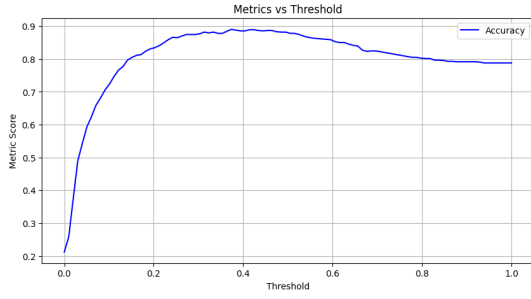


Fig. 4. Performance of RF model across varying classification thresholds.

TABLE II
PERFORMANCE COMPARISON OF MODELS ON OVARIAN CANCER RISK PREDICTION BASED ON OVA NOTE1 (THRESHOLD = 0.5, K = 5-FOLD CV)

Model	Accuracy	AUC	MAE	MSE	R^2	Running time (s)
XGBoost	0.885	0.646	0.152	0.054	0.257	95
XGBoost + ADASYN	0.869	0.713	0.218	0.080	-0.093	149
RF	0.882	0.624	0.152	0.054	0.266	128
RF + ADASYN	0.887	0.740	0.218	0.078	-0.073	265
LinearR	0.899	0.640	0.157	0.051	0.308	65
LinearR + ADASYN	0.884	0.691	0.234	0.078	-0.066	94
BERT	0.889	0.711	0.128	0.044	0.450	653
ViHealthBERT	0.861	0.695	0.152	0.056	0.307	1256

can classify malignancy from clinical free-text with moderate to high accuracy. While ML models achieved high accuracy (up to 0.882), their AUC values were lower (0.624–0.740), indicating limited discriminatory ability. BERT achieved a comparative accuracy, suggesting better capacity for capturing contextual cues from unstructured text. However, language models required significantly longer training time (up to 1256s), which may limit their practicality in resource-constrained settings. In general, language models demonstrate potential for direct text-based classification, but still face challenges in clinical concept grounding and generalization, as shown in the interpretability example in Fig. 5.

C. Prediction/Classification of structured data based models using OvaNOTE2 dataset

In this sub-section, we compare the performance of models trained with structured data from the OvaNOTE2 dataset. We evaluate both the regression models (output the probability) and the classification model (output the class). It answers the second question: *Q2 - What is the model that best fits the structured data?*

TABLE III
PERFORMANCE COMPARISON OF MODELS ON OVARIAN CANCER RISK PREDICTION BASED ON OVA NOTE2 (THRESHOLD = 0.5, K = 5-FOLD CV)

Model	Accuracy	AUC	MAE	MSE	R^2	Running time (s)
XGBoost	0.915	0.938	0.112	0.034	0.583	15
XGBoost + ADASYN	0.916	0.931	0.125	0.038	0.530	17
RF	0.926	0.932	0.105	0.032	0.609	15
RF + ADASYN	0.926	0.930	0.110	0.036	0.559	17
LinearR	0.901	0.921	0.131	0.041	0.493	15
LinearR + ADASYN	0.888	0.922	0.178	0.060	0.260	17
TabPFN	0.924	0.926	0.103	0.033	0.592	4126
TabPFN + ADASYN	0.927	0.935	0.095	0.031	0.618	1885

In Tab. III, among them, RF and TabPFN showed the best overall performance. RF achieved high accuracy and low error with very fast training time. TabPFN, a transformer-based model, also performed well but required significantly more computation. Fig. 6 shows a comparison of actual vs. predicted benign/malignancy scores by RF on Fold 3. While RF captured general trends, it tended to underperform on high-risk cases, with prediction errors varying across subgroups. Applying ADASYN slightly improved accuracy but sometimes introduced noise, especially in simpler models. Overall, RF proved highly effective for structured clinical data, offering a strong balance between performance, speed, and interpretability, often outperforming more complex deep learning models in this setting.

Fig. 7 presents a comparative analysis of the RF model’s performance on the OvaNOTE1 and OvaNOTE2 datasets using five evaluation metrics: Accuracy, AUC, MAE, MSE, and R^2 . The model demonstrates consistently better performance on OvaNOTE2 across all metrics. Notably, the AUC and R^2 values on OvaNOTE2 are significantly higher than those on OvaNOTE1, indicating improved discriminative ability and stronger predictive power on the second dataset. Additionally, lower MAE and MSE scores on OvaNOTE2 suggest enhanced precision in risk prediction. These results imply that the data characteristics or quality in OvaNOTE2 may be more conducive to learning, highlighting the importance of dataset composition in model evaluation.

Figure 5 compares LIME explanations for Patient 2239 with Benign Probability equals 22.8% or Malignant Risk equals 77.2%. The structured-data model correctly identified malignancy, assigning a low benign probability at around 0.36, while the text-based model incorrectly predicted benign with 0.95 confidence. From the left panel, the text-based model heavily relied on generic or ambiguous terms, which do not clearly signal malignancy. These terms appear frequently across both benign and malignant reports, leading to confusion. Notably, malignant-specific phrases were given minimal importance. In contrast, the structured model in the right panel focused on clinically validated features: a large solid diameter (>0.26), presence of papillae, and absence of benign indicators (e.g., no pelvic fluid, no >10 cysts). These features contributed strongly to the malignancy prediction and aligned with domain knowledge, indicating a more robust reasoning process. This case highlights that while LLMs are useful for unstructured data, structured models offer better accuracy and clinical alignment, especially in complex malignant cases.

V. CONCLUSIONS

In this study, we proposed and evaluated an AI-based pipeline for ovarian cancer benign probability prediction using ultrasound medical notes, integrated OCR, acronym decoding, and LLM-assisted information extraction to generate structured data. We evaluated this pipeline for predicting ovarian tumor malignancy risk using ultrasound medical notes from 805 Vietnamese patients. We compared traditional ML/DL models. Results showed that while BERT models captured

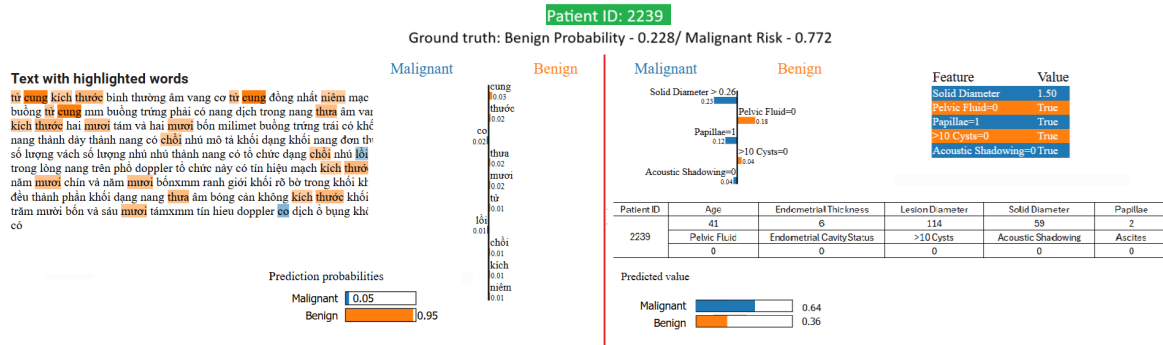


Fig. 5. RF interpretability visualization using LIME for Patient ID 2239.

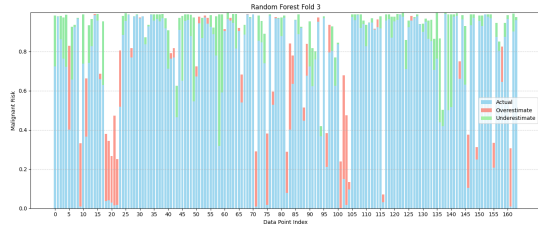


Fig. 6. RF Prediction Gaps and Benign Probability (Fold 4).

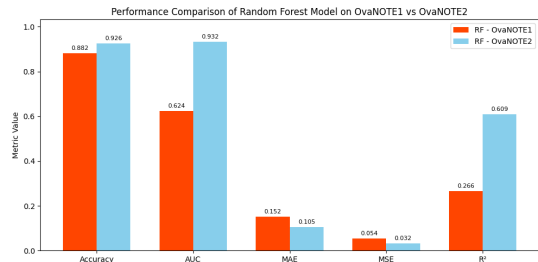


Fig. 7. RF Performance on OvaNOTE1 vs OvaNOTE2

semantic information from clinical narratives, they were limited in predictive performance and interpretability, particularly in handling domain-specific ambiguity. In contrast, structured ML models, especially RF, consistently outperformed LLM-based approaches in both regression and classification tasks. In particular, tabular models showed stronger alignment with clinical reasoning and exhibited higher robustness when diagnosing malignant cases. Visualization results further revealed that LLMs can be misled by generic terms in text, whereas structured features contributed more directly to accurate prediction. These findings emphasize the continued importance of structured clinical data and domain-driven feature engineering in developing reliable, interpretable AI tools for cancer diagnosis in low-resource environments.

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